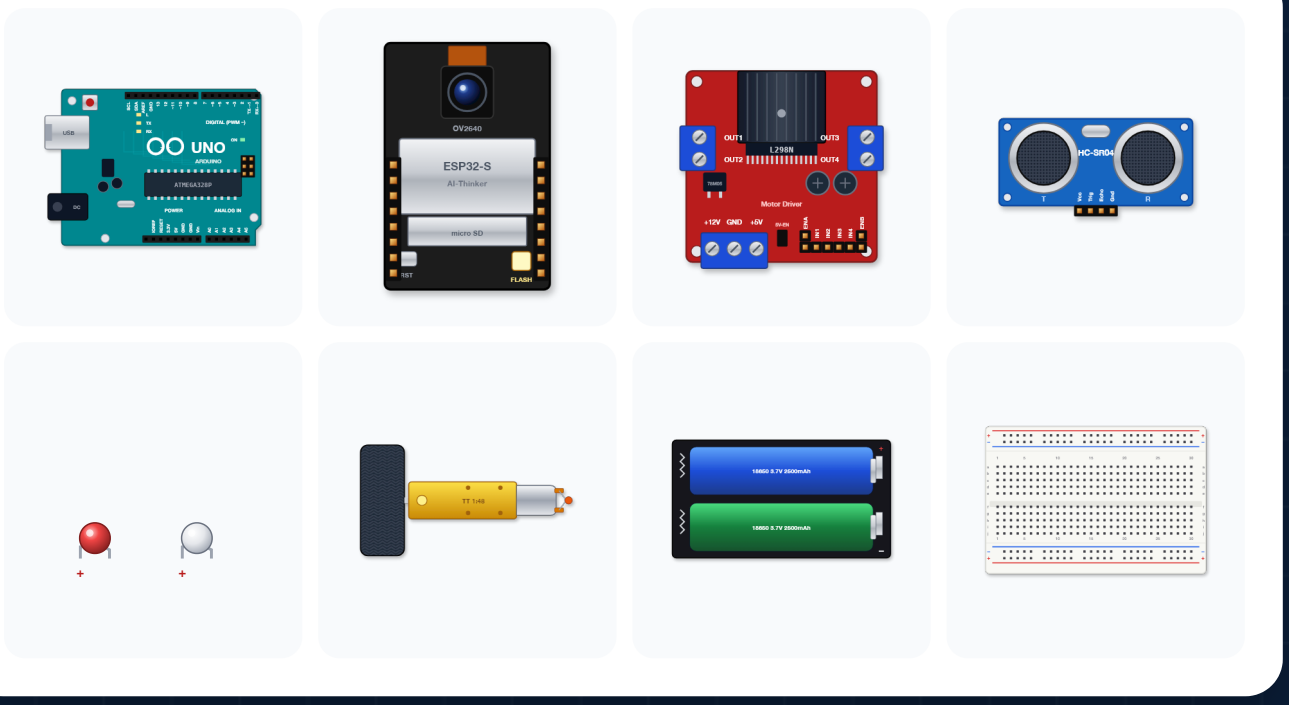
ROBOT CAR PROJECT

MODULE 01

01

Overview & Parts List

What we are building, how it works, and exactly which parts to buy.



Name _____ Class / Group _____ Date _____

4WD · Arduino UNO · ESP32-CAM
Edition 2026

MODULE 01

1 Project overview

In this project you build a **four-wheel-drive robot car that you drive from a smartphone**. A camera at the front sends live video to the phone. A sensor at the back measures the distance to obstacles. When you reverse, the car warns you like a real car's **parking sensor**: a buzzer beeps and a red light flashes, faster and faster as you get closer, and the car brakes by itself before it hits anything. The phone shows the measurement as a **radar**.

LEARNING OBJECTIVES · ABOUT 45 MIN

By the end of this module you can:

- Explain the job of every part in the car and how they talk to each other
- Choose correct parts (voltage, current, type) and avoid common bad buys
- Read a power budget and explain why the car uses two power rails (7.4 V and 5 V)
- Check your kit against the parts list before you start building

What the finished car can do

Phone control
Joystick in the phone browser. No app to install.

Live video
The camera at the front streams to the phone.

Parking alarm
Buzzer and LED beep faster as an obstacle gets closer.

Auto-brake
The car refuses to reverse into an obstacle.

Radar display
Rear distance shown as a radar on the phone.

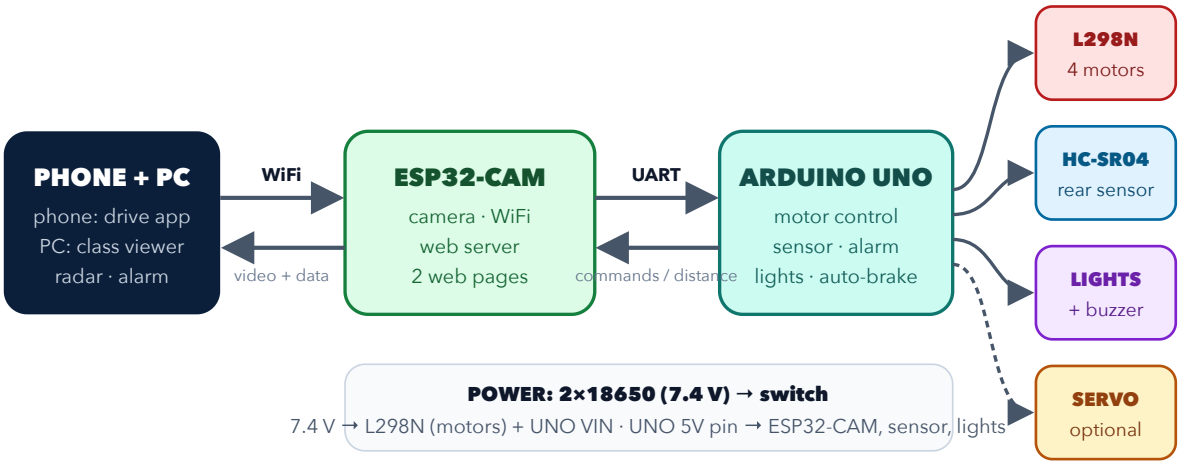
Headlights
Two white front lights: OFF, ON or HAZARD from the app.

Classroom viewer
A second page for a PC or projector: video, radar, chart, status.

Photos & video
Save photos on the phone, record video on the PC.

Fail-safe
Stops by itself if WiFi or the phone is lost.

How it works



Why two small computers?

The **ESP32-CAM** is great at WiFi and video, but it has only about 4 free pins and is busy streaming. The **Arduino UNO** handles the motors, the sensor and the alarm in real time. If WiFi lags, the parking alarm and auto-brake still react instantly.

Two power rails

7.4 V from the batteries goes to the motor driver and to the UNO's **VIN** pin. The UNO's own regulator makes the **5 V** for all the electronics. A big **2200 µF capacitor** at the ESP32-CAM covers the short dips when the motors start, so the camera does not reboot.

MODULE 01

2 Parts list

Everything below is needed for the standard build. Prices are approximate (online shops, 2026) and only for planning.



4WD chassis kit

2 acrylic plates, 4 × TT gear motors (3–6 V, 1:48), 4 wheels 65 mm, spacers, screws

Qty 1 kit — $12-18



Arduino UNO R3 (or Nano)

ATmega328P, USB cable. CH340 clones are fine

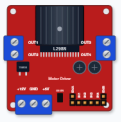
Qty 1 — $5-12



ESP32-CAM (AI-Thinker)

OV2640 camera, 4 MB PSRAM. Get the ESP32-CAM-MB USB board too

Qty 1 + MB — $9-13



L298N motor driver

Dual H-bridge with heatsink, 5 V regulator, ENA/ENB jumpers

Qty 1 — $2-4



HC-SR04 ultrasonic

5 V, 2–400 cm. Mounted at the rear of the car

Qty 1 — $1-2



2 × 18650 + holder

Protected Li-ion 3.7 V, real 2000–3000 mAh, series holder = 7.4 V

Qty 2 + 1 — $8-14



Half breadboard

400 points with + / – rails on both sides

Qty 1 — $1-2



Rocker switch

≥ 3 A, often included in the kit

Qty 1 — $0.3



Active buzzer 5 V

ACTIVE type: it beeps on plain 5 V (test it on a battery)

Qty 1 — $0.5



LEDs 5 mm

1 red (rear warning) + 3 white (reverse light + 2 headlights). Buy a few spares

Qty 4 — $0.4



Resistors

220 Ω × 4 · 1 kΩ × 1 · 2 kΩ × 1 (or 2 × 1 kΩ)

Qty 6 — $0.5

Small parts

Part	Specification	Qty	Used for
Electrolytic capacitor	**2200 µF 16 V** (any value 1000–2200 µF, 10 V or more)	1	Stops the ESP32-CAM rebooting when motors start
Ceramic capacitor	100 nF (marked "104")	4	Soldered across each motor: less electrical noise
Jumper wires	Dupont 20 cm: M-M, M-F, F-F (20 of each)	1 set	Signal connections
Power wire	22 AWG (0.5 mm²) red + black, ~1 m each	–	Battery → switch → L298N, L298N → UNO VIN
ESP32-CAM-MB	Micro-USB programmer board for the ESP32-CAM	1	Uploading code (or use an FTDI adapter)
18650 charger	2-slot Li-ion charger (or 2 × TP4056)	1	Charging the batteries safely

Avoid these bad buys

- **"9900 mAh" 18650 cells**: they are fakes that sag under load and make the ESP32 reboot. Buy known brands (Samsung, LG, Sony/Murata, Molicel).
- **Passive buzzer**: it only clicks on DC. You need an **active** buzzer, which beeps when connected straight to 5 V.
- **ESP32-CAM without PSRAM**: the video then works only at a small size. Look for "4 MB PSRAM".
- **9 V PP3 battery or 4 × AA**: far too weak for 4 motors.

Checklist: check your kit

- [ ] 4 TT motors + 4 wheels + chassis plates
- [ ] Arduino UNO + USB cable
- [ ] ESP32-CAM + camera + MB board
- [ ] L298N motor driver
- [ ] HC-SR04 sensor
- [ ] Active buzzer
- [ ] 2 × 18650 cells (charged) + holder
- [ ] Switch, breadboard, jumper wires
- [ ] LEDs: 1 red + 3 white
- [ ] Resistors 220 Ω ×4, 1 kΩ, 2 kΩ
- [ ] Capacitors 2200 µF + 4 × 100 nF
- [ ] Multimeter + soldering iron

Checked by (student)

Approved by (teacher)

MODULE 01

3 Tools & power budget

Tools

Tool	Needed for	
Multimeter	Checking the battery polarity and the 5 V rail	REQUIRED
Soldering iron + solder	Motor wires and capacitors	REQUIRED
Small screwdrivers	L298N screw terminals	REQUIRED
Wire stripper / cutter	Preparing power wires	REQUIRED
Computer + Arduino IDE 2	Programming both boards	REQUIRED
FTDI USB-serial adapter	Only if you have no ESP32-CAM-MB board	OPTIONAL
Hot glue, zip ties, M3 nylon standoffs	Neat, vibration-proof mounting	OPTIONAL

Power budget: why these parts?

Load	Rail	Typical	Peak
4 × TT motors	7.4 V via L298N	0.6–1.0 A	2.5 A (start / stall)
ESP32-CAM (WiFi + video)	5 V	180 mA	**350 mA spikes**
Arduino UNO itself	5 V	50 mA	60 mA
HC-SR04 + buzzer	5 V	30 mA	45 mA
4 LEDs (rear, reverse, 2 headlights)	5 V	0–60 mA	60 mA
Total from the UNO 5V pin		**≈ 0.3 A**	**≈ 0.5 A** for a moment: OK for the UNO regulator
SG90 servo (optional)	**own 5 V supply**	100 mA	600 mA: too much for the UNO, never on this rail

Why 7.4 V?
The L298N loses about 2 V, so the TT motors receive about **5.4 V**, which is their sweet spot. Two 18650 cells in series give exactly that.

Why the UNO's 5 V, not the L298N's?
The L298N's small regulator gets hot and its 5 V dips when the motors start, so the ESP32-CAM resets ("brownout"). The UNO's regulator is stronger. With the 2200 µF capacitor it keeps the camera running. It gets warm: that is normal.